Mansoura University
Faculty of Computers and Information
Total Grade: Max. 60 marks
Date: 22/6/2025 (3 pages)

Final Exam
Model (1)

Academic Year: 2024/2025
Course Name: Decision Support Systems.
Grade: 3rd
Time Allowed: 120 minutes.

Q.I) Choose the right answers. (50 marks)

1. ………. approach would be appropriate for a decision maker who seeks to achieve the best results if the worst happens

A) Minimax	B) Regret	C) Maximax	D) Maximin

2. When the decision-maker wants to avoid risk and ensure the best possible outcome under the worst-case scenario, the decision maker use ………..approach.

A) EVPI	B) regret	C) optimistic	D) conservative

3. Which level of management is primarily supported by EIS?

A) operational	B) administrative	C) management	D) strategic

4. Decision-making under uncertainty is more difficult because of availability of information.

A) TRUE	B) FALSE

5. In decision-making under risk, probabilities are:

A) Unknown	B) Assumed to be equal	C) Known and quantifiable	D) Ignored

6. What is the first step in the decision-making process?

A) Execute the decision	B) Identify the purpose of the decision	C) Gather information	D) Evaluate alternatives

7. ……… is the increase in the expected profit if the decision maker knows with certainty which state of nature occur.

A) AHP	B) EV	C) EVPI	D) ROMC

8. ………is a financial tool that helps businesses determine the sales volume or revenue needed to cover all costs associated with producing and selling a product or service.

A) AHP model	B) Pro forma	C) Break even	D) Ratio

9. AHP model combines qualitative and quantitative criteria?

A) TRUE	B) FALSE

10. Which tool helps prioritize causes of a problem?

A) Break-even analysis	B) Pareto Chart	C) Pro forma statements	D) Sensitivity analysis

11. Which analysis assumes stable conditions?

A) Dynamic	B) Sensitivity	C) Static	D) Optimistic

12. …………… refers to the technologies, tools, and processes used to collect, analyze, and present business data.

A) DSS	B) OLTP	C) Business intelligence	D) OLAP

13. The decision-making process can be defined as the activity that leads to the solution of a decision-making problem involving at most ………. Alternatives.

A) One	B) Two	C) Three	D) none of them

14. ……… is a phase of the DSS process that consists of discovering, identifying, and understanding the problems occurring in the organization.

A) Intelligence	B) Design	C) Control	D) Implementation

15. Decision-making under certainty assumes:

A) Multiple uncertain outcomes	B) Probabilities are known	C) Perfect knowledge of outcomes	D) No alternatives exist

16. If a company has fixed costs of $100,000, variable costs of $1000 per unit, and a selling price of $3000 per unit, this means that the company must sell …………, units to cover all its costs (BEA).

A) 1000	B) 2000	C) 50	D) 5

17. ………… analysis can summarize the anticipated financial results for a specific period in the future

A) AHP model	B) Pro forma	C) Break even	D) Ratio

18. Which management level uses DSS for resource allocation decisions?			
A) Strategic	B) Operational	C) Tactical	D) Executive
19. According to figure 1, the decision maker will recommend buying ....based on AHP method.			
A) Escort	B) Miata	C) Civic	D) Saturn
20. What is the Consistency Ratio (CR) threshold in AHP?			
A) $CR > 0.5$	B) $CR > 0.1$	C) $CR < 0.01$	D) $CR = 1.0$
21. In machine learning, what is a performance measure?			
A) A method to increase speed	B) A criterion to evaluate performance	C) A specific programming language	D) A data storage solution
22. In an online learning system, the system may take longer to adapt to new data if the learning rate is ----.			
A) too low	B) moderate	C) too high	D) none of these
23. What is the effect of reducing max_depth in a decision tree?			
A) Lowers accuracy	B) Reduces overfitting	C) No effect	D) Increases complexity
24. Which of the following statements is true about hyperparameters?			
A) They automatically adjust during training	B) They remain constant during training	C) They are parameters of the model	D) They must be set after training begins
25. What is the primary output of a machine learning system?			
A) Data	B) Rules	C) Program	D) Hardware
26. What does the experience (E) refer to in the machine learning definition?			
A) The results of the model's predictions	B) User's interactions with the software	C) The training data provided to the model	D) The hardware specifications
27. In reinforcement learning, what does the term "policy" refer to?			
A) A strategy for maximizing rewards	B) A set of labels for the training data	C) A dimensionality reduction method	D) An algorithm for clustering
28. Which type of machine learning is characterized by labeled training data			
A) Supervised	B) Unsupervised	C) Semisupervised	D) Reinforcement
29. --------- learning is a training method based on rewarding desired behaviors and/or punishing undesired ones.			
A) Supervised	B) Unsupervised	C) Semisupervised	D) Reinforcement
30. Which of the following algorithms is commonly used for clustering in unsupervised learning?			
A) KNN	B) SVM	C) Linear regression	D) K-Means
31. Which learning method is more suitable for continuous streams of data, like stock prices?			
A) Batch learning	B) Offline learning	C) Online learning	D) None of these
32. Which type of learning is generally slower and requires more resources for training?			
A) Batch learning	B) Continuous learning	C) Online learning	D) None of these
33. Which approach is NOT part of feature engineering?			
A) Feature selection	B) Feature extraction	C) Data cleaning	D) None of these
34. What is it called when a model performs well on the training data but does not generalize well?			
A) Overfitting	B) Underfitting	C) Fitting	D) None of these
35. What regression metric is best if the training data has many outliers?			
A) MSE	B) RMSE	C) MAE	D) None of these
36. Using OvO strategy, -------------- binary classifiers are needed if the number of classes is 5.			
A) 5	B) 10	C) 15	D) 20
37. What is the ratio of positive instances that are correctly detected by the classifier?			
A) Accuracy	B) Precision	C) Recall	D) F1-score
38. Which criterion does a decision tree use to evaluate the best split when training?			
A) Accuracy	B) Variance reduction	C) Gini impurity	D) None of these
39. What is the time complexity of making a prediction with a Decision Tree?			
A) $O(m)$	B) $O(m^2)$	C) $O(m\log_2(m))$	D) $O(\log_2(m))$
40. Which node attribute represents the number of training instances of each class that this node applies to?			
A) Samples	B) Value	C) Gini	D) Class

Q.2) Answer the following question (10 marks).

According to **TABLE 1** (profit payoff table), determine which alternative the decision maker will select according to the optimistic, pessimistic, and regret approaches. If the decision maker knows that the probabilities of the states of nature are p(s1)=0.5, p(s2)=0.1, p(s3)=0.4, what is the best alternative using the expect value approach using **DT** and determine the **EVPI**. Write the required steps.

Table 1: Profit payoff

		States of nature		
		S1	**S2**	**S3**
Alternatives	**A1**	18	1	12
	A2	10	9	14
	A3	13	6	9
	A4	5	4	2

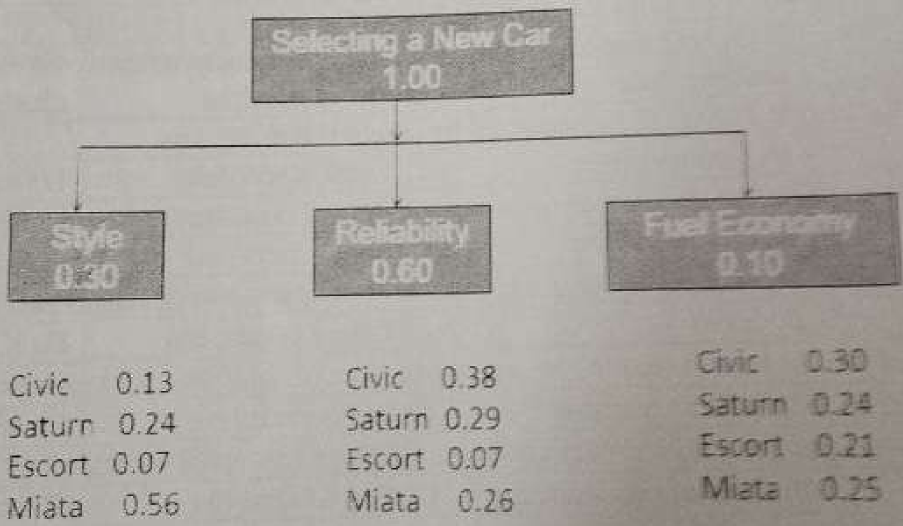

Figure 1

Good Luck.